YiRage (YPK) — RuntimeFusion Architecture

Inference engine, RuntimeFusion, search plan, backend manager, implementations, and hardware

Dashed frame = architecture zone
Solid frame = subsystem
Box = module and responsibility

PROJECT OVERVIEW

YiRage is a multi-backend LLM kernel compiler and runtime centered on graph superoptimization and RuntimeFusion.

The architecture separates the inference engine, RuntimeFusion, search plans, the backend manager, backend implementations, and hardware targets.

INFERENCE ENGINE · scheduling owner

- **vLLM** — continuous batch · PagedAttention
- **SGLang** — RadixAttention · extend meta
- **Decode Metadata** — block_table · seq_lens · radix hit · SM budget · batch

RuntimeFusion · RF.step(inputs, meta)

- **RF.step** — select 0..N capsules per decode step
- **FusionCapsule** — MLP gate_up + down · yirage.core
- **FusionCapsule** — segment pipeline · gate_up + down
- **Native Fallback** — engine native operators

SEARCH & PLAN · offline / warm-up

- **superoptimize** — KN / TB graph search
- **FusionPlan** — µGraph · profile + fingerprint
- **MuGraph cache** — shape bucket restore
- **Ray Coordinator** — optional distributed search
- **Serving Tiers** — seed_verify · full_tb · full_tb_ray

PYTHON API & BACKEND MANAGER

- **KNGraph / TBGraph** — kernel transpiler
- **BackendRegistry** — search + execute · same backend
- **yirage.core** — Cython runtime
- **yirage.serving** — RF · Capsule · cert
- **MACA Config** — warp = 64 · mxcc
- **CPU MLIR JIT** — optional

BACKEND IMPLEMENTATIONS

- **CUDA** — nvcc · warp 32
- **MACA** — mxcc · warp 64
- **CPU** — OpenMP · MKL
- **MPS** — Apple GPU
- **Ascend** — CANN
- **ROCm** — AMD GPU
- **Triton** — DSL kernels
- **+ extra** — TPU · XPU · FPGA · MLIR

HARDWARE TARGETS

- **NVIDIA GPU**
- **MetaX C500/C550**
- **x86 / ARM CPU**
- **Apple Silicon**
- **Huawei Ascend**
- **AWS Neuron** — Intel XPU

The engine owns scheduling and KV. RuntimeFusion owns FusionCapsules. BackendRegistry binds search and execution to the same backend.